Responsive Webpage sprint report

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1 Objective

The primary objectiveof this sprint was to develop a fully responsive, modern web page utilizing HTML, Tailwind CSS, and vanilla JavaScript, incorporating advanced features and following best practices for web development.

2 Key Functionalities

2.1 Dark Mode Toggle

Key Features

- Consistent theme preference using localstorage
- Tailwind's dark: variant used for styling consistency

2.2 Form Validation

Key Features

- Real-time validation as users type
- Email format validation using regex patterns
- Required field checking
- Prevention of form submission if validation fails

Technical Implementation

Event listeners on input and blur events trigger validation functions. Custom validation logic checks field requirements and displays appropriate feedback. The form prevents submission if validation fails, ensuring data integrity.

2.3 Animated Statistics Counter

Key Features

- Intersection Observer API triggers animation when counters enter viewport
- Animation triggers only once per page load for performance

2.4 Performance Optimization

Lazy Loading

Lazy loading image technique that defers loading of images off-screen until they're needed, significantly improves initial page load performance.

```
<img width="800" height="600" src="resources/images/jpg/icon1.jpg"
   alt="Icon 1" class="rounded-sm mx-auto" loading="lazy">
```

- Setting explicit width and height prevents layout shifts (CLS) by reserving space for the image before it loads.

Image Optimization with Modern Formats

To improve performance and follow modern web best practices, image optimization techniques were implemented using the < *picture* > element and multiple image formats.

e.g.

```
<picture class="flex justify-center">
    <source srcset="resources/images/avif/icon2-400.avif 400w, resources/
       images/avif/icon2-800.avif 800w, resources/images/avif/icon2-1200.
       avif 1200w" type="image/avif">

    <source srcset="resources/imageswebp/icon2.webp" type="image/webp">

    <source srcset="resources/images/jpg/icon2.jpg" type="image/jpg">

    <img src="resources/images/jpg/icon2.jpg" alt="Icon 2" class="rounded-
       sm text-center mx-auto" loading="lazy" width="800" height="600">
</picture>
```

Key Features

- Use of AVIF and WebP formats for better compression and reduced file size
- PNG/JPG fallback for older browser compatibility
- Responsive images using srcset
- Lazy loading with loading="lazy"
- Explicit width and height attributes to prevent CLS (Cumulative Layout Shift)

SVG Icons

- Replaced fontawesome cdn icons with **SVG** format
- Faster rendering and improved performance

2.5 Accessibility and SEO Enhancements

Accessibility Improvements

- Use of semantic HTML elements (header, nav, main, section)
- Implementation of **ARIA attributes** where necessary
- Improved screen reader support

```
<button id="mobile-menu-toggle"
        aria-label="Toggle menu">
</button>
```

- The **aria-label="Toggle menu"** ensures screen readers clearly describe the button's function

SEO Optimization

- Proper meta tags (description, viewport, charset, og, twitter..)
- Semantic document structure

3 Deployment and Performance Evaluation

3.1 Project Deployment

The project has been successfully deployed using **Vercel**.

Live Application: `https://sprint2-webpage.vercel.app`

GitHub Repository: `https://github.com/a-naitdayaa/sprint2-webpage.git`

3.2 Lighthouse Performance Testing

To evaluate performance, accessibility, best practices, and SEO, the application was tested using Google Lighthouse.

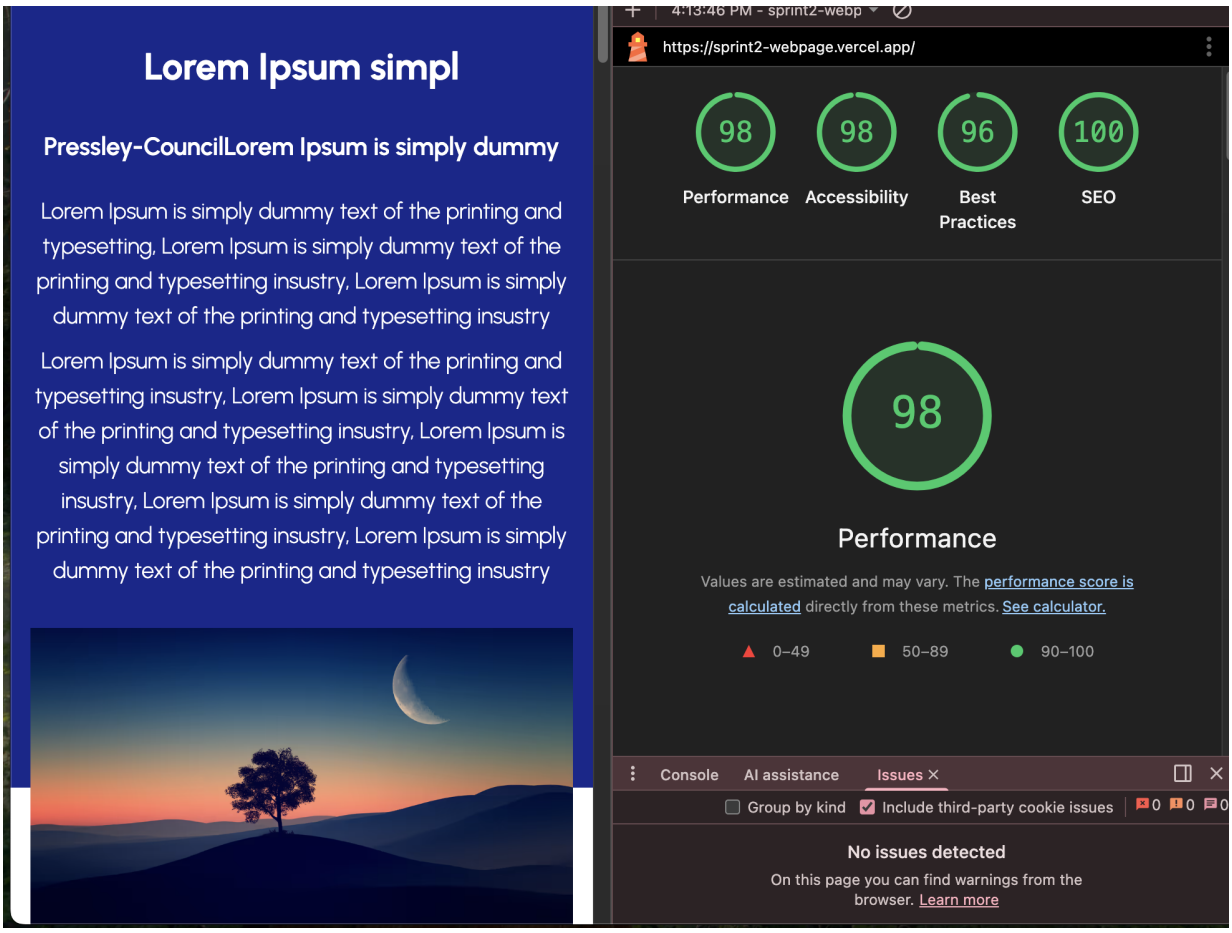

Figure 1: Lighthouse Performance Report